



General Use SOP

Compressed Gases

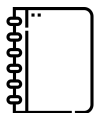
Process or Experiment Description

This standard operating procedure (SOP) provides general guidance working safely with compressed gases. If a gas has other hazardous properties (e.g., it is flammable, oxidizing, [toxic](#), etc.), contact EH&S before use, as special controls may be needed. For questions about the applicability of any item in this SOP, contact your Principal Investigator/Supervisor or EH&S at 650-723-0448.

Hazards

Compressed gases are asphyxiants (except compressed air and oxygen) and have inherent pressure hazards. Compressed gases may also be flammable, toxic, and/or corrosive. Some gases (e.g., silane, diborane, phosphine) are pyrophoric, meaning they will ignite spontaneously in air. All compressed gases expand significantly in volume when released to air, which may result in pressure build-up, oxygen displacement, or heat generation. Common compressed gases include nitrogen, carbon dioxide, helium, and argon.

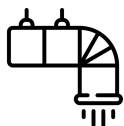
Control of Hazards



General

- Check connections and hoses regularly for [leaks](#) using a specific monitoring instrument or soapy water (or equivalent). You can also leak test by pressurizing your system for 1 hour and checking if it maintains the same pressure; if pressure drops, there is a leak. If your system uses a toxic, flammable, or reactive gas, perform these checks with a non-toxic and non-reactive gas (preferably compressed air) instead.
- Verify that all system components including tubing, valves, etc. are chemically compatible with your gas and are rated for the pressure you intend to use.
- Do not alter or modify compressed gas cylinders.

- Replace valve caps when cylinders are not in use and before moving them.
- Remove damaged or defective cylinders from service (contact the cylinder vendor for assistance).
- Toxic gases are regulated by Santa Clara County and may require special safeguards. See EH&S's page on [Toxic Gas](#) for more information.
- For gas systems involving more than a few simple components, develop a specific SOP that includes clear diagrams and/or pictures of the system clearly labeling all components. This is particularly recommended for reactive and/or toxic gases.
- Piping and tubing containing compressed gases must be labeled every 20 feet, at every significant change in direction, and at every wall penetration with the name of the gas and the direction of flow.
- If your lab has a personal or area gas sensor, maintain it according to the manufacturer's instructions, including periodic calibration, bump testing, and replacement. Consult with EH&S to assess sensor necessity and selection.



Engineering/Ventilation Controls

Work in a well-ventilated area, such as a wet lab. If the process doesn't allow working with compressed gases in a well-ventilated area, contact Environmental Health and Safety at 650-723-0448



Personal Protective Equipment

In addition to proper street clothing (long pants or equivalent that covers legs and ankles, and close-toed non-perforated shoes that completely cover the feet), wear the following Personal Protective Equipment (PPE):

- Safety glasses
- Lab coat



Special Handling Procedures and Storage Requirements

Safe Handling:

- Compressed gas cylinders must be transported using hand trucks or other appropriate means.

Never transport unsecured compressed gas cylinders.

- Transport cylinders in an upright position.
- Do not ride in an elevator with compressed gas cylinders. One person should send the elevator while another person receives it.

Safe Storage:

- Store cylinders in an upright position.
- Cylinders should be capped when not in use.
- Secure compressed gas cylinders >26 inches tall to an anchored rack using two metal chains (at 1/3 and 2/3 cylinder height). Cylinders <26 inches tall only require one chain to be secured.
- No more than two cylinders may be secured with one pair of chains.
- Ensure that the cylinder label is legible and includes the identity and concentration of all components.
- Clearly mark cylinders as full, in use, or empty.
- Store compressed gas cylinders away from heat sources.
- Segregate compressed gases according to hazard class and chemical compatibility. Ensure separation of flammable and oxidizing gases.
- Store flammable gases away from ignition sources (including unprotected electrical connections) and oxidizing gases.
- Follow all substance-specific storage guidance provided in SDS.

Gas Release and Accident Procedures

All incidents should be reported to EH&S with an [incident report form](#).

In the event of a gas leak, shut off the gas at the source if it is safe to do so, then evacuate the area. Call 911 for health-threatening leaks (e.g., a toxic, flammable, or corrosive gas leak, or if pressure could build and cause an explosion).

A person who was exposed to a compressed gas (besides compressed air or oxygen) should get to fresh air as soon as possible. If they were exposed to a corrosive or toxic gas, they should seek medical attention immediately.

Refer to EH&S guidance on [hazardous materials incidents](#) for general procedures for incident response.

Those who work with hazardous chemicals are entitled to receive medical attention/consultation when:

- A spill, leak, explosion or other occurrence results in a hazardous exposure (potential

overexposure).

- Symptoms or signs of exposure to a hazardous chemical develop.

Waste Disposal

Coordinate with the appropriate vendor for return of cylinders. If the vendor will not take the cylinder, contact EH&S for disposal.

Minimum Training Requirements

- General Safety & Emergency Preparedness (EHS-4200)
- Chemical Safety for Laboratories (EHS-1900)
- Compressed Gas Safety (EHS-2200)
- Laboratory-specific training

Approval Required

Consult with your PI regarding the need for prior approval. Laboratory personnel shall seek and the PI must provide prior approval of any chemical usage involving [Restricted Chemicals or High Risk Procedures](#).

Designated Area

For compressed gases that are also considered [particularly hazardous substances](#) (i.e., select carcinogens, reproductive toxins, and highly acutely toxic chemicals), a designated area shall be established per the other applicable SOP.